



Fig. 3: Install alsa-utils



Fig. 4: Open module file in nano editor

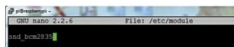


Fig. 5: Add 'snd_bcm2835' in the file



Fig. 6: Open mplayer configuration file

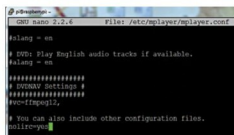


Fig. 7: Add 'nolirc=yes' to the file

install all the software. Refer 'Set up Network for Raspberry Pi' published in May 2013 issue for getting the network connection up on your Raspberry Pi. Once done, you can either connect a keyboard and a display to Raspberry Pi and start following the installations using LxTerminal or you can access Raspberry Pi remotely using SSH and execute all the commands directly.

Festival text-to-speech installation. The Festival Speech Synthesis System is a general-purpose multi-lingual speech synthesis system used here to direct the visually challenged person to take the right path. Festival is designed to support multiple language-

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.

es, and comes with support for English (British and American pronunciation), Welsh and Spanish. Voice packages exist for several other languages, such as Castilian Spanish, Czech, Finnish, Hindi, Italian, Marathi, Polish, Russian and Telugu.

First update and upgrade the Raspberry Pi related software using the commands below and reboot your Raspberry Pi for a fresh look:

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

To install sound on Raspberry Pi, you need to install 'alsa' sound utilities using the command below. Refer Fig. 3.

```
$ sudo apt-get install alsa-utils
```

Now edit the module file at '/etc/modules' using nano editor. For that, run the command below to open the file in nano editor and add 'snd_bcm2835.' Refer Fig. 4 and Fig. 5.

```
$ sudo nano /etc/module
```

If 'snd_bcm2835' line is already present, leave the file as it is. Once done, save the file by 'CTRL+O' and exit by 'CTRL+X.'

Now install the mplayer audio/movie player by command mentioned below:

```
$ sudo apt-get install mplayer
```

To sort out the mplayer error message, edit file '/etc/mplayer/mplayer.conf' using nano editor and add

'nolirc=yes.' Refer Fig. 6 and Fig. 7.

```
$ sudo nano /etc/mplayer/mplayer.conf
```

Once you have completed all the steps mentioned above, you can install Festival text-to-speech using the command below:

```
$ sudo apt-get install festival
```

You can try out Festival installation using the command below in the terminal and you will hear 'Hello EFY' in the earphones.

```
$ echo "Hello EFY" | festival --tts
```

Python GPIO module for Raspberry Pi. The code is written in Python programming language and compiled using Python in the Raspberry Pi. Python comes as a built-in package for the Raspberry Pi and we only need to install the GPIO package modules.

Download the 'RPi.GPIO-0.1.0.tar.gz' from '<http://pypi.python.org/packages/source/R/RPi.GPIO/RPi.GPIO-0.1.0.tar.gz>' and extract into a suitable directory using the command:

```
$ tar -zxvf RPi.GPIO-0.1.0.tar.gz
```

Now use 'cd' command to go inside the extracted directory and install the package using:

```
$ sudo python setup.py install
```

Once the package is installed, download the source code in a suitable directory and run it using:

```
$ sudo python filename.py
```

You can also copy the code in nano editor and save it with .py extension. Please note that Python is strictly indented language and the indentation normally change when you paste the code in nano editor. ●

The author is a student of M.Sc. at IIT, Kharagpur

Read more stories on Power Electronics at

www.electronicb2b.com

Log on to www.electronicb2b.com and be in touch with the Electronics B2B Fraternity 24x7

Top Battery Stories

- ☐ Electronics in power sector: Challenges & opportunities
- ☐ The latest in home UPS systems
- ☐ The latest in industrial inverters
- ☐ UPS and inverter market evoked mixed response in 2012-13: While it was smooth sailing for the giants, SMEs found the going tough
- ☐ Industrial UPS systems are becoming highly customised
- ☐ The latest in switch mode power supplies

ELECTRONICS

B2B
INDUSTRY IS AT A

CLICK OF A BUTTON

